Kaniuar Bacho

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"I am passionate about discovering and understanding the unknown with all of my heart."

Education

M.Sc. IT Security, Ruhr University Bochum (RUB), *Germany* Apr 2022 - Apr 2024

- Cumulative GPA: 96% (excellent)
- Master's Thesis: "Classical Verification of Quantum Computations".
  Advisor: Prof. Dr. Michael Walter
- Special focus on Quantum Cryptography and Post-Quantum Cryptography.
- Selected Coursework:
  - Cryptography Introduction I + II (*Spring & Fall 2022, both grade 100%*)
  - Cryptography (*Fall 2022, grade 100%*)
  - Cryptographic Protocols (*Spring 2023, grade 98%*)
  - Post-Quantum Cryptanalysis (*Spring 2023, grade 100%*)
  - Quantum Cryptography (*Fall 2022, grade 100%*)
  - Quantum Circuits (*Spring 2023, grade 100%*)
  - Quantum Information and Computation (*Fall 2022, grade 93%*)
  - Quantum Algorithms Seminar (*Spring 2023*)
  - Zero-Knowledge Proof Systems (*Spring 2023*)
- Computer Security Ranking 2019: Horst Görtz Institute (HGI) in Bochum rank 1 in Europe.
  Rank 13 worldwide in terms of publications at top computer security conferences.

B.Sc. Mathematics, University of Bonn, *Germany* 2017 - 2021

- Cumulative GPA: A (very good)
- Special focus on Algebraic Number Theory and Algebraic Geometry. Minor in Psychology.
- Bachelor's Thesis: "Classification of Irreducible Representations of $\mathrm{GL}_2(\mathbb{F}_{p^n})$ over Algebraically Closed Fields".
  Advisors: Dr. Johannes Anschütz & Prof. Dr. Peter Scholze (*grade A+*)
- ShanghaiRanking's Global Ranking of Academic Subjects 2020: Bonn rank 13 worldwide in mathematics.
  University of Bonn has two Fields medallists: Gerd Faltings & Peter Scholze

Research and Work Experience

Teaching Assistant, *Ruhr University Bochum (RUB) & University of Bonn* 2019 - 2024

- Conducted weekly instructional sessions, assessed assignments, and evaluated exams.
- Quantum Information and Computation (Prof. Michael Walter), *Fall 2023*.
- Cryptographic Protocols (Prof. Eike Kiltz), *Spring 2023*.
- Analysis II (Prof. Margherita Disertori), *Spring 2019*. Graded A+ by the professor and my students.

Quantum Technologies Intern, *TÜV Information Technology, Essen* Sep 2022 - Oct 2022

- Advised companies on securing their Quantum Key Distribution (QKD) setups against attack vectors.
- Gave department-wide presentation on BB84 and contemporary side-channel attacks on QKD devices.

Volunteer Mentor for Freshmen, *University of Bonn* Oct 2019 - Oct 2020

- Offered tailored guidance, sharing effective learning strategies and enhancing problem-solving skills.

Max Planck Institute (MPI) for Mathematics, Visiting Researcher, *Bonn* Sep 2018 - Oct 2018

- Research on an open problem in Algebraic Topology in cooperation with other national mathematics winners.

Selected Honors & Awards

German Academic Scholarship Foundation (Studienstiftung des deutschen Volkes) 2017 - 2024
- Top ∼0.5% students in/from Germany are awarded. Most prestigious scholarship foundation.
- Selection is based on excellent academic records, social engagements, rhetorical and social skills, academic and sport competitions.
- Awarded full funding for my Bachelor's and Master's studies.

International Olympiad in Cryptography (NSUCRYPTO) - Silver Medal, Ranked 3rd 2023

International Mathematical Olympiad (IMO) German Selection Round (4 times) 2014 - 2017

'German Mathematical Society' Member (DMV) 2017

'German National Mathematics Competition' Two-time National Winner (BWM) 2016 & 2017
- Top ∼10 of ∼1500 participants will be crowned national winners. Most prestigious mathematical competition in Germany alongside the German Mathematical Olympiad.
- Solving Algebra, Number Theory, Geometry, Combinatorics problems over three rounds (see my solutions).
- In the last round, I talked with a prominent expert in Arithmetic Geometry, assessing how quickly one can comprehend abstract structures and tackle complex problems. Winners are selected based on these criteria.

German TV Show 'WDR' Interview 2017
- Interview about the achievement of winning the aforementioned mathematical competition.

German Mathematical Olympiad (German MO) 2010 - 2016
- Top ∼200 of the ∼200.000 participants will advance to the finals. First prize in third round needed.
- I was a team member representing the federated state of NRW for the finals in 2016.

Westphalia Boxing Champion & Vice NRW Boxing Champion 2015
- The national championship comprises four rounds, with each round containing a maximum of three fights. To progress to the next round, one must emerge victorious in all fights.
- Successfully won round two and secured a second-place position in round three.

Projects

Web Development (Personal Website) 2023

Implementing the 'McEliece Post-Quantum-Cryptosystem' in SageMath 2023

Hacking a Virtual Bank 2023
- In the four-month university project, we explored hacking a virtual bank using techniques such as identifying logical flaws, XSS, SQL injections, local file inclusion, remote code execution, and CSRF.

Bitcoin Wallet Cluster 2022
- Used the 'Bitcoin Blockchain Explorer API' to gather generated blocks, extracting public keys from wallets. These keys were then clustered together using various heuristics to associate them with individual users.

Talks

Slides of my talks are available at kaniuarbacho.github.io

Quantum Random Walks
- Quantum Algorithms Seminar at Ruhr University Bochum (*Jul 2023*).

Quantum Key Distribution
- Department-wide presentation at company 'TÜV Information Technology' in Essen (*Oct 2022*).

Quadratic Forms over $\mathbb{Q}_p$
- Algebraic Number Theory Seminar at University of Bonn (*Nov 2020*).

Classification of Irreducible Representations of $\mathrm{GL}_2(\mathbb{F}_{p^n})$
- Defense of the Bachelor's Thesis at University of Bonn (*Sep 2020*).

Representation Theory of Lie Algebras

- Lie Algebras Seminar at University of Bonn (*May 2020*).

Random Walks on the Symmetric Group

- Markov Chains and Stochastic Algorithms Seminar at University of Bonn (*Jul 2018*).

Service

Corrector at the German Mathematical Olympiad 2016 (national round)

About Me

Programming	Python, SageMath, C/C++, HTML/CSS, Bash (Linux), Assembly, Git, LaTeX.
Languages	German (native), Kurdish (native), English (proficient, TOEFL Score 103).
Hobbies	Reading inspirational and self-improvement books, Gym, Cardistry.